

EX-NO:04

MONGODB CRUD OPERATORS

27-10-2023

AIM:

To Create a document-based database using MongoDB and manipulate the data.

CREATE A COLLECTIONS:

```
> db.createCollection("Location")
< { ok: 1 }
Locations>
```

INSERTONE:

```
> db.Location.insertOne({Id:110, Name:"Rose", Age:20, Location_Id:1020, Street_Add:"348 TNHB Ave", City:"Seoul", Country_Id:"SK", Country:"South Korea"})
< {
  acknowledged: true,
  insertedId: ObjectId("65439765cacc66bdfa5b9692")
}
```

```
_id: ObjectId('65439765cacc66bdfa5b9692')
Id: 110
Name: "Rose"
Age: 20
Location_Id: 1020
Street_Add: "348 TNHB Ave"
City: "Seoul"
Country_Id: "SK"
Country: "South Korea"
```

INSERTMANY:

```
> db.Location.insertMany([
  {Id:116, Name:"Shri", Age:21, Location_Id:2600, Street_Add:"9702 Chester Road", City:"Stretford", Country_Id:"UK", Country:"United Kingdom"},
  {Id:117, Name:"Lisa", Age:21, Location_Id:2700, Street_Add:"Rua Frei Caneca 1360", City:"Sao Paulo", Country_Id:"BR", Country:"Brazil"}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId("65439a08d9e349ee8580691e"),
    '1': ObjectId("65439a08d9e349ee8580691f")
  }
}
```

```

_id: ObjectId('65439a08d9e349ee8580691e')
Id: 116
Name: "Shri"
Age: 21
Location_Id: 2600
Street_Add: "9702 Chester Road"
City: "Stretford"
Country_Id: "UK"
Country: "United Kingdom"

```

```

_id: ObjectId('65439a08d9e349ee8580691f')
Id: 117
Name: "Lisa"
Age: 21
Location_Id: 2700
Street_Add: "Rua Frei Caneca 1360"
City: "Sao Paulo"
Country_Id: "BR"
Country: "Brazil"

```

FIND:

```
> db.Location.find({Age:20})
```

```

< {
  _id: ObjectId("654396c4dadd8160e6a8c250"),
  Id: 100,
  Name: 'Steven',
  Age: 20,
  Location_Id: 1000,
  Street_Add: '1297 Via Cola di Rie',
  City: 'Roma',
  Country_Id: 'IT',
  Country: 'Italy'
}
{
  _id: ObjectId("654396c4dadd8160e6a8c254"),
  Id: 104,
  Name: 'Bruce',
  Age: 20,
  Location_Id: 1400,
  Street_Add: '2014 Jabberwocky Rd',
  City: 'Southlake',
  Country_Id: 'US',
  Country: 'United States of America'
}

```

```

{
  _id: ObjectId("65439765cacc66bdfa5b9692"),
  Id: 110,
  Name: 'Rose',
  Age: 20,
  Location_Id: 1020,
  Street_Add: '348 TNH B Ave',
  City: 'Seoul',
  Country_Id: 'SK',
  Country: 'South Korea'
}

```

FINDONE:

```
> db.Location.findOne({Name:"Steven"})
< {
  _id: ObjectId("654396c4dadd8160e6a8c250"),
  Id: 100,
  Name: 'Steven',
  Age: 20,
  Location_Id: 1000,
  Street_Add: '1297 Via Cola di Rie',
  City: 'Roma',
  Country_Id: 'IT',
  Country: 'Italy'
}
```

USING PROJECTION IN FIND AND FINDONE:**INCLUDING SPECIFIC FIELDS:**

```
> db.Location.find({ Age:20 }, { Name:1, Age:1 })
< {
  _id: ObjectId("654396c4dadd8160e6a8c250"),
  Name: 'Steven',
  Age: 20
}
{
  _id: ObjectId("654396c4dadd8160e6a8c254"),
  Name: 'Bruce',
  Age: 20
}
{
  _id: ObjectId("65439765cacc66bdfa5b9692"),
  Name: 'Rose',
  Age: 20
}
```

EXCLUDING SPECIFIC FIELDS:

```
> db.Location.find({ Age:20 }, { Country:0, _id:0 })
< {
  Id: 100,
  Name: 'Steven',
  Age: 20,
  Location_Id: 1000,
  Street_Add: '1297 Via Cola di Rie',
  City: 'Roma',
  Country_Id: 'IT'
}
```

```
{
  Id: 104,
  Name: 'Bruce',
  Age: 20,
  Location_Id: 1400,
  Street_Add: '2014 Jabberwocky Rd',
  City: 'Southlake',
  Country_Id: 'US'
}
{
  Id: 110,
  Name: 'Rose',
  Age: 20,
  Location_Id: 1020,
  Street_Add: '348 TNHBAve',
  City: 'Seoul',
  Country_Id: 'SK'
}
```

USING FINDONE WITH PROJECTION:

```
> db.Location.findOne({ Age:20 }, { Name:1, City:1 })
< {
  _id: ObjectId("654396c4dadd8160e6a8c250"),
  Name: 'Steven',
  City: 'Roma'
}
```

UPDATEONE:

```

_id: ObjectId('654396c4dadd8160e6a8c250')
Id: 100
Name: "Steven"
Age: 20
Location_Id: 1000
Street_Add: "1297 Via Cola di Rie"
City: "Roma"
Country_Id: "IT"
Country: "Italy"

```

```

> db.Location.updateOne({ Name: "Steven" }, { $set: { Age: 21, City: "Milan" } })
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

```

_id: ObjectId('654396c4dadd8160e6a8c250')
Id: 100
Name: "Steven"
Age: 21
Location_Id: 1000
Street_Add: "1297 Via Cola di Rie"
City: "Milan"
Country_Id: "IT"
Country: "Italy"

```

```

> db.Location.updateOne({ Name: "Steven" }, { $inc: { Age: 2 } })
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

```

_id: ObjectId('654396c4dadd8160e6a8c250')
Id: 100
Name: "Steven"
Age: 23
Location_Id: 1000
Street_Add: "1297 Via Cola di Rie"
City: "Milan"
Country_Id: "IT"
Country: "Italy"

```

UPDATEMANY:

```

_id: ObjectId('654396c4dadd8160e6a8c254')
Id: 104
Name: "Bruce"
Age: 20
Location_Id: 1400
Street_Add: "2014 Jabberwocky Rd"
City: "Southlake"
Country_Id: "US"
Country: "United States of America"

```

```

_id: ObjectId('65439765cacc66bdfa5b9692')
Id: 110
Name: "Rose"
Age: 20
Location_Id: 1020
Street_Add: "348 TNHB Ave"
City: "Seoul"
Country_Id: "SK"
Country: "South Korea"

```

```

> db.Location.updateMany({ Age: 20 }, { $set: { Country: "Spain" } })
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}

```

```

_id: ObjectId('654396c4dadd8160e6a8c254')
Id: 104
Name: "Bruce"
Age: 20
Location_Id: 1400
Street_Add: "2014 Jabberwocky Rd"
City: "Southlake"
Country_Id: "US"
Country: "Spain"

```

```

_id: ObjectId('65439765cacc66bdfa5b9692')
Id: 110
Name: "Rose"
Age: 20
Location_Id: 1020
Street_Add: "348 TNHB Ave"
City: "Seoul"
Country_Id: "SK"
Country: "Spain"

```

```

_id: ObjectId('654396c4dadd8160e6a8c252')
Id: 102
Name: "Lex"
Age: 23
Location_Id: 1200
Street_Add: "2017 Shinjuku-ku"
City: "Tokyo"
Country_Id: "JP"
Country: "Japan"

```

```

_id: ObjectId('654396c4dadd8160e6a8c253')
Id: 103
Name: "Alexander"
Age: 24
Location_Id: 1300
Street_Add: "9450 Kamiya-cho"
City: "Hiroshima"
Country_Id: "JP"
Country: "Japan"

```

```

> db.Location.updateMany({ Country: "Japan" }, { $inc: { Age: 1 } })
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}

```

```

_id: ObjectId('654396c4dadd8160e6a8c252')
Id: 102
Name: "Lex"
Age: 24
Location_Id: 1200
Street_Add: "2017 Shinjuku-ku"
City: "Tokyo"
Country_Id: "JP"
Country: "Japan"

```

```

_id: ObjectId('654396c4dadd8160e6a8c253')
Id: 103
Name: "Alexander"
Age: 25
Location_Id: 1300
Street_Add: "9450 Kamiya-cho"
City: "Hiroshima"
Country_Id: "JP"
Country: "Japan"

```

REPLACEONE:

```

_id: ObjectId('654396c4dadd8160e6a8c251')
Id: 101
Name: "Neena"
Age: 22
Location_Id: 1100
Street_Add: "93091 Calle della Testa"
City: "Venice"
Country_Id: "IT"
Country: "Italy"

```

```

> db.Location.replaceOne({ Name: "Neena" }, { Name: "Neena", Age: 25, City: "Madrid", Country: "Spain" })
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

```

_id: ObjectId('654396c4dadd8160e6a8c251')
Name: "Neena"
Age: 25
City: "Madrid"
Country: "Spain"

```

DELETEONE:

```

> db.Location.deleteOne({ Name: "Neena" })
< {
  acknowledged: true,
  deletedCount: 1
}

```

DELETEMANY:

```

> db.Location.deleteMany({ Age: 20 })
< {
  acknowledged: true,
  deletedCount: 2
}

```

COMPARISON:**EQUAL:**

```
> db.Location.find({ Age: { $eq: 25 } })
< {
  _id: ObjectId("654396c4dadd8160e6a8c253"),
  Id: 103,
  Name: 'Alexander',
  Age: 25,
  Location_Id: 1300,
  Street_Add: '9450 Kamiya-cho',
  City: 'Hiroshima',
  Country_Id: 'JP',
  Country: 'Japan'
}
```

```
> db.Location.find({ Country: { $eq: "Italy" } })
< {
  _id: ObjectId("654396c4dadd8160e6a8c250"),
  Id: 100,
  Name: 'Steven',
  Age: 23,
  Location_Id: 1000,
  Street_Add: '1297 Via Cola di Rie',
  City: 'Milan',
  Country_Id: 'IT',
  Country: 'Italy'
}
```

GREATER THAN:

```
> db.Location.find({ Age: { $gt: 25 } })
< {
  _id: ObjectId("654396c4dadd8160e6a8c255"),
  Id: 105,
  Name: 'David',
  Age: 29,
  Location_Id: 1500,
  Street_Add: '2011 Interiors Blvd',
  City: 'South San Francisco',
  Country_Id: 'US',
  Country: 'United States of America'
}
```



```

{
  _id: ObjectId("654396c4dadd8160e6a8c25c"),
  Id: 112,
  Name: 'Jose Manuel',
  Age: 26,
  Location_Id: 2200,
  Street_Add: '12-98 Victoria Street',
  City: 'Sydney',
  Country_Id: 'AU',
  Country: 'Australia'
}
{
  _id: ObjectId("654396c4dadd8160e6a8c25f"),
  Id: 115,
  Name: 'Shelli',
  Age: 26,
  Location_Id: 2500,
  Street_Add: 'Magdalen Centre, The Oxford Science Park',
  City: 'Oxford',
  Country_Id: 'UK',
  Country: 'United Kingdom'
}

```

LESS THAN:

```

> db.Location.find({ Age: { $lte: 19 } })
< {
  _id: ObjectId("654396c4dadd8160e6a8c256"),
  Id: 106,
  Name: 'Valli',
  Age: 18,
  Location_Id: 1600,
  Street_Add: '2007 Zagora St',
  City: 'South Brunswick',
  Country_Id: 'Us',
  Country: 'United States of America'
}

```

```

{
  _id: ObjectId("654396c4dadd8160e6a8c25b"),
  Id: 111,
  Name: 'Ismael',
  Age: 19,
  Location_Id: 2100,
  Street_Add: '1298 Vileparle€',
  City: 'Bombay',
  Country_Id: 'IN',
  Country: 'India'
}
{
  _id: ObjectId("654396c4dadd8160e6a8c25e"),
  Id: 114,
  Name: 'Den',
  Age: 19,
  Location_Id: 2400,
  Street_Add: '8204 Arthur St',
  City: 'London',
  Country_Id: 'UK',
  Country: 'United Kingdom'
}

```

LOGICAL:

AND:

```

> db.Location.find({$and: [{ Age: 25 }, { Country: "Japan" } ]})
< {
  _id: ObjectId("654396c4dadd8160e6a8c253"),
  Id: 103,
  Name: 'Alexander',
  Age: 25,
  Location_Id: 1300,
  Street_Add: '9450 Kamiya-cho',
  City: 'Hiroshima',
  Country_Id: 'JP',
  Country: 'Japan'
}

```

OR:

```
> db.Location.find({$or: [{ Age: 25 }, { Country: "Italy" } ]})
< {
  _id: ObjectId("654396c4dadd8160e6a8c250"),
  Id: 100,
  Name: 'Steven',
  Age: 23,
  Location_Id: 1000,
  Street_Add: '1297 Via Cola di Rie',
  City: 'Milan',
  Country_Id: 'IT',
  Country: 'Italy'
}
{
  _id: ObjectId("654396c4dadd8160e6a8c253"),
  Id: 103,
  Name: 'Alexander',
  Age: 25,
  Location_Id: 1300,
  Street_Add: '9450 Kamiya-cho',
  City: 'Hiroshima',
  Country_Id: 'JP',
  Country: 'Japan'
}
```

NOT:

```
> db.Location.find({Age: {$not: { $eq: 25 } } })
< {
  _id: ObjectId("654396c4dadd8160e6a8c250"),
  Id: 100,
  Name: 'Steven',
  Age: 23,
  Location_Id: 1000,
  Street_Add: '1297 Via Cola di Rie',
  City: 'Milan',
  Country_Id: 'IT',
  Country: 'Italy'
}
{
  _id: ObjectId("65439a08d9e349ee8580691f"),
  Id: 117,
  Name: 'Lisa',
  Age: 21,
  Location_Id: 2700,
  Street_Add: 'Rua Frei Caneca 1360',
  City: 'Sao Paulo',
  Country_Id: 'BR',
  Country: 'Brazil'
}
```

AGGREGATIONS:

GROUPING AND COUNTING:

```
> db.Location.aggregate([ { $group: { _id: "$Country", count: { $sum: 1 } } } ] )
< {
  _id: 'China',
  count: 1
}
{
  _id: 'Australia',
  count: 1
}
{
  _id: 'Singapore',
  count: 1
}
```

```
{
  _id: 'United Kingdom',
  count: 3
}
{
  _id: 'India',
  count: 1
}
{
  _id: 'Brazil',
  count: 1
}
{
  _id: 'United States of America',
  count: 3
}
{
  _id: 'Canada',
  count: 2
}
{
  _id: 'Italy',
  count: 1
}
```

```
{
  _id: 'Japan',
  count: 2
}
```

FILTERING AND PROJECTING FIELDS:

```
> db.Location.aggregate([{$match: { Country_Id: "US" }},{ $project: { Name: 1, Age: 1, _id: 0 } }])
< {
  Name: 'David',
  Age: 29
}
{
  Name: 'Diana',
  Age: 21
}
```

ADDFIELDS:

```
> db.Location.aggregate([{$addFields: {FullAddress: { $concat: ["$Street_Add", ", ", "$City", ", ", "$Country"] }} }])
< {
  _id: ObjectId("654396c4dadd8160e6a8c250"),
  Id: 100,
  Name: 'Steven',
  Age: 23,
  Location_Id: 1000,
  Street_Add: '1297 Via Cola di Rie',
  City: 'Milan',
  Country_Id: 'IT',
  Country: 'Italy',
  FullAddress: '1297 Via Cola di Rie, Milan, Italy'
}
```

```
{
  _id: ObjectId("654396c4dadd8160e6a8c252"),
  Id: 102,
  Name: 'Lex',
  Age: 24,
  Location_Id: 1200,
  Street_Add: '2017 Shinjuku-ku',
  City: 'Tokyo',
  Country_Id: 'JP',
  Country: 'Japan',
  FullAddress: '2017 Shinjuku-ku, Tokyo, Japan'
}
```

```
{
  _id: ObjectId("654396c4dadd8160e6a8c253"),
  Id: 103,
  Name: 'Alexander',
  Age: 25,
  Location_Id: 1300,
  Street_Add: '9450 Kamiya-cho',
  City: 'Hiroshima',
  Country_Id: 'JP',
  Country: 'Japan',
  FullAddress: '9450 Kamiya-cho, Hiroshima, Japan'
}
```

```
{
  _id: ObjectId("65439a08d9e349ee8580691e"),
  Id: 116,
  Name: 'Shri',
  Age: 21,
  Location_Id: 2600,
  Street_Add: '9702 Chester Road',
  City: 'Stretford',
  Country_Id: 'UK',
  Country: 'United Kingdom',
  FullAddress: '9702 Chester Road, Stretford, United Kingdom'
}
```

```
{
  _id: ObjectId("65439a08d9e349ee8580691f"),
  Id: 117,
  Name: 'Lisa',
  Age: 21,
  Location_Id: 2700,
  Street_Add: 'Rua Frei Caneca 1360',
  City: 'Sao Paulo',
  Country_Id: 'BR',
  Country: 'Brazil',
  FullAddress: 'Rua Frei Caneca 1360, Sao Paulo, Brazil'
}
```

RESULT:

Thus, Created a document-based database using MongoDB and performed the crud operations.

OBSERVATION	RECORD	SIGNATURE